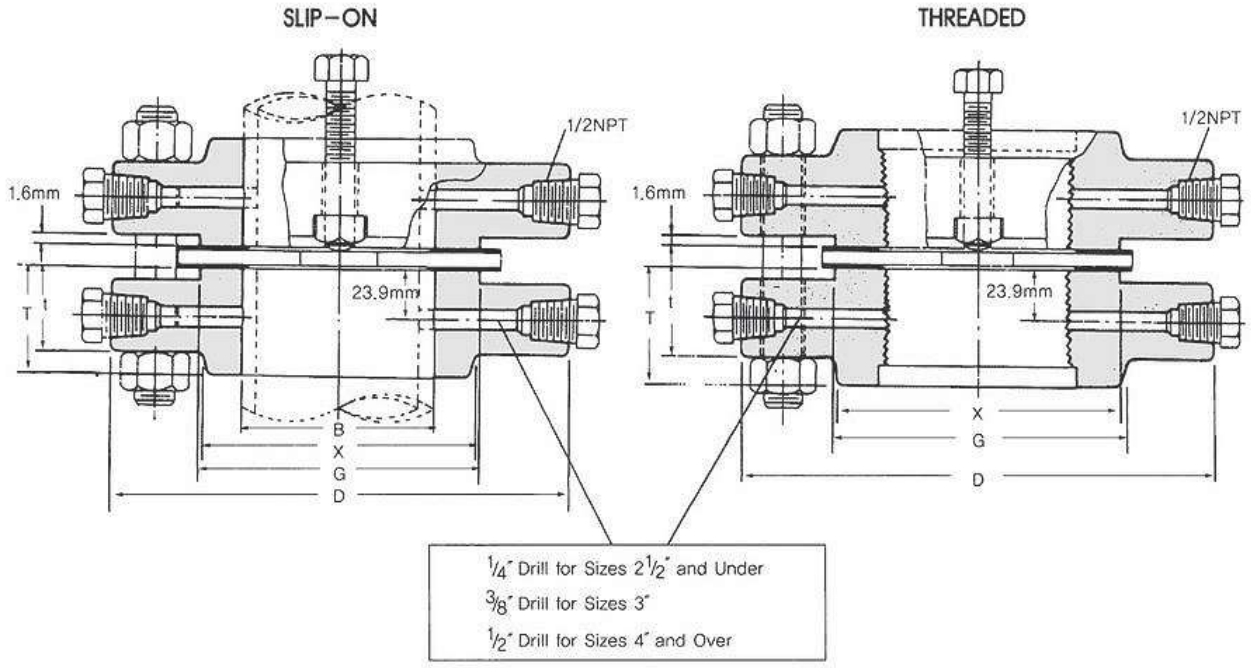


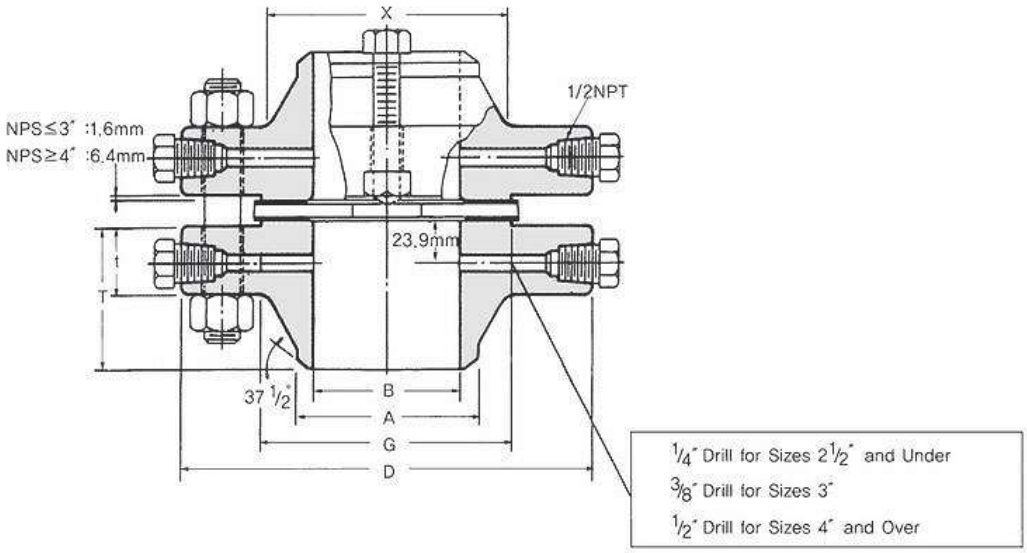
Unit:mm

Nominal Pipe Size	Pitch Diam.of Ring and Groove	Ring Number	DEPTH OF JACK SCREW SLOT	JACK SCREW SIZE	DRILLING TEMPLATE				
			Raised Face	Raised Face	Diam. of Bolt Circle	Number of Bults	Diam. of Stud Bolts (inch)	Diam. of Bolt Holes	Length of Stud Bolts
	P								Raised Face
1	50.8	R16	9.7	Jack screw size for 1" thru 24" those shown for length and diameter of bolts.	88.9	4	5/8	17.5	139.7
1 1/4	60.3	R18	9.7		98.6	4	5/8	17.5	152.4
1 1/2	68.3	R20	12.7		114.3	4	3/4	20.6	152.4
2	82.6	R23	9.7		127.0	8	5/8	17.5	152.4
2 1/2	101.6	R26	12.7		149.4	8	3/4	20.6	152.4
3	123.8	R31	12.7		168.1	8	3/4	20.6	152.4
4	149.2	R37	12.7		200.2	8	3/4	20.6	152.4
5	181.0	R41	12.7		235.0	8	3/4	22.4	152.4
6	211.1	R45	12.7		269.7	12	3/4	22.4	152.4
8	269.9	R49	15.7		330.2	12	7/8	25.4	158.8
10	323.9	R53	19.1		387.4	16	1	28.4	165.1
12	381.0	R57	22.4		450.9	16	1 1/8	31.8	177.8
14	419.1	R61	22.4		514.4	20	1 1/8	31.8	184.2
16	469.9	R65	25.4		571.5	20	1 1/4	35.1	196.9
18	533.4	R69	25.4		628.7	24	1 1/4	35.1	203.2
20	584.2	R73	25.4		685.8	24	1 1/4	35.1	215.9
24	692.2	R77	31.8		812.8	24	1 1/2	41.1	241.3

(5) Unless otherwise specified, unions of 1" (25.4mm) thru 24" (609.6mm) furnished with carbon steel regular square headed bolts with semifinished American Standard heavy series hex nuts.

CLASS 400 ORIFICE FLANGES

WELDING NECK (RAISED FACE)



ASME B16.36 FORGED FLANGES

Unit:mm

Nominal Pipe Size	Outside Diam.of Flange	THICKNESS OF FLANGE(t)		Diam. of Hub at Base	Diam. of Raised Face	Diam. of Hub at Bevel	LENGTH THRU HUB(T)				BORE(B)	
							Welding Neck		Slip-on & Threaded		Welding Neck	Slip-on
	D	Raised Face	Ring Joint	X	G	A	Raised Face	Ring Joint	Raised Face	Ring Joint		
1	124	38.1	31.8	53.8	50.8	33.5	82.6	76.2	47.8	41.1	See Note(1) To be specified by purchaser	34.5
1 1/4	133	38.1	31.8	63.5	63.5	42.2	84.1	77.7	46.0	39.6		43.2
1 1/2	155	38.1	31.8	69.9	73.2	48.3	85.9	79.2	47.8	41.1		49.5
2	165	38.1	31.8	84.1	91.9	60.5	85.9	79.2	49.3	42.9		62.0
2 1/2	191	38.1	31.8	100.1	104.6	73.2	88.9	82.6	50.8	44.5		74.7
3	210	38.1	31.8	117.3	127.0	88.9	88.9	82.6	52.3	46.0		90.7
4	254	35.1	35.1	146.1	157.2	114.3	88.9	88.9	50.8	50.8		116.1
5	279	38.1	38.1	177.8	185.7	141.2	101.6	101.6	53.8	53.8		143.8
6	318	41.1	41.1	206.2	215.9	168.4	103.1	103.1	57.2	57.2		170.7
8	381	47.8	47.8	260.4	269.7	219.2	117.3	117.3	68.3	68.3		221.5
10	445	53.8	53.8	320.5	323.9	273.1	124.0	124.0	73.2	73.2		276.4
12	521	57.2	57.2	374.7	381.0	323.9	136.7	136.7	79.2	79.2		327.2
14	584	60.7	60.5	425.5	412.8	355.6	149.4	149.4				359.2
16	648	63.5	63.5	482.6	469.9	406.4	152.4	152.4				410.5
18	711	66.5	66.5	533.4	533.4	457.2	165.1	165.1				461.8
20	775	69.9	69.9	587.2	584.2	508.0	168.1	168.1				513.1
24	914	76.2	76.2	701.5	692.2	609.6	174.8	174.8				564.4

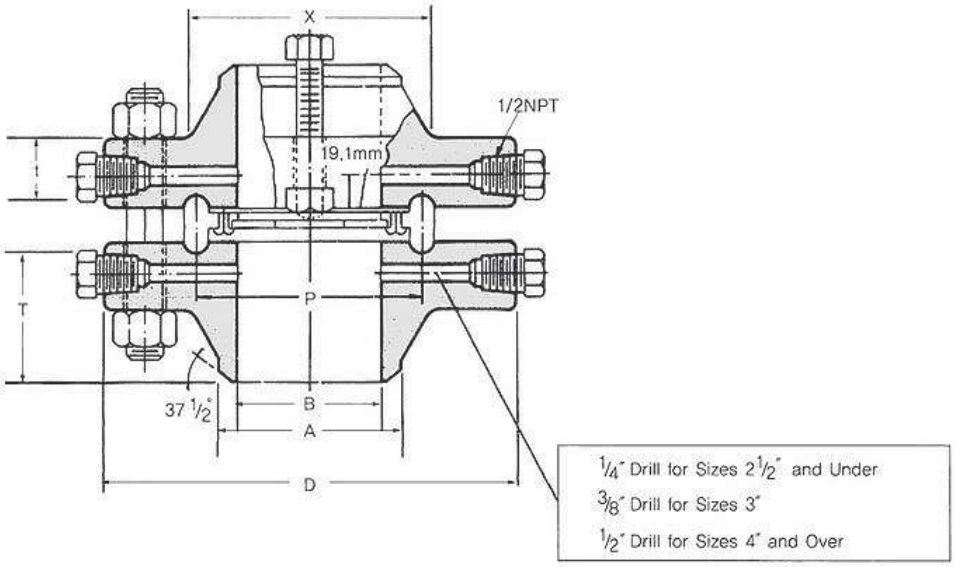
Notes

(1) For the inside diameter of pipes (corresponding to 'Bore' (B) of Welding Neck Flanges), refer to page 15.

(2) Class 400 Flanges of sizes 3" (76.2mm) and smaller will be furnished with 0.06" (1.6mm) raised face, which is included in 'Thickness' (t) and 'Lengththrough Hub'(T)
The 0.25" (6.35mm) raised face for sizes 4" (101.6mm) and larger is not included in (t) and (T).

(3) Each union includes two carbon steel jack screw bolts with hex nuts.

WELDING NECK (RING–JOINT TYPE)



Unit:mm

Nominal Pipe Size	Pitch Diam.of Ring and Groove	Ring Number	DEPTH OFJACK SCREW SLOT		JACK SCREW SIZE		DRILLING TEMPLATE					
			Raised Face	Ring Joint	Raised Face (inch)	Ring Joint (inch)	Diam. of Bolt Circle	Number of Bolt	Diam. of Stud Bolts (inch)	Diam. of Bolt Holes	Length of Stud Bolts	
	P										Raised Face	Ring Joint
1	50.8	R16	9.7	6.4	5/8 X 4.00	5/8 X 4.75	88.9	4	5/8	17.5	127.0	146.1
1 1/4	60.3	R18	9.7	6.4	5/8 X 4.00	5/8 X 4.75	98.6	4	5/8	17.5	127.0	120.7
1 1/2	68.3	R20	12.7	6.4	3/4 X 4.25	3/4 X 5.00	114.3	4	3/4	21.0	133.4	152.4
2	82.6	R23	9.7	6.4	5/8 X 4.00	5/8 X 4.75	127.0	8	5/8	17.5	127.0	152.4
2 1/2	101.6	R26	12.7	6.4	3/4 X 4.25	3/4 X 5.00	149.4	8	3/4	20.6	133.4	158.8
3	123.8	R31	12.7	6.4	3/4 X 4.25	3/4 X 5.00	168.1	8	3/4	20.6	133.4	158.8
4	149.2	R37	6.4	15.7	3/4 X 3.00	3/4 X 4.00	200.2	8	7/8	25.4	139.7	152.4
5	181.0	R41	6.4	15.7	3/4 X 3.00	3/4 X 4.00	235.0	8	7/8	25.4	146.1	158.8
6	211.1	R45	12.7	22.4	1X 3.50	1X 4.00	269.7	12	7/8	25.4	158.8	165.1
8	269.9	R49	12.7	22.4	1X 3.50	1X 4.50	330.2	12	1	28.4	171.5	184.2
10	323.9	R53	12.7	22.4	1X 4.00	1X 4.50	387.4	16	1 1/8	31.8	190.5	203.2
12	381.0	R57	12.7	22.4	1X 4.00	1X 5.00	450.9	16	1 1/4	35.1	203.2	215.9
14	419.1	R61	12.7	22.4	1X 4.25	1X 5.00	514.4	20	1 1/4	35.1	209.6	228.6
16	469.9	R65	12.7	22.4	1X 4.25	1X 5.00	571.5	20	1 3/8	38.1	222.3	235.0
18	533.4	R69	12.7	22.4	1X 4.50	1X 5.00	628.7	24	1 3/8	38.1	235.0	241.3
20	584.2	R73	12.7	22.4	1X 4.75	1X 5.50	685.8	24	1 1/2	41.1	247.7	260.4
24	692.2	R77	12.7	22.4	1X 5.00	1X 6.00	812.8	24	1 3/4	47.8	279.4	285.8

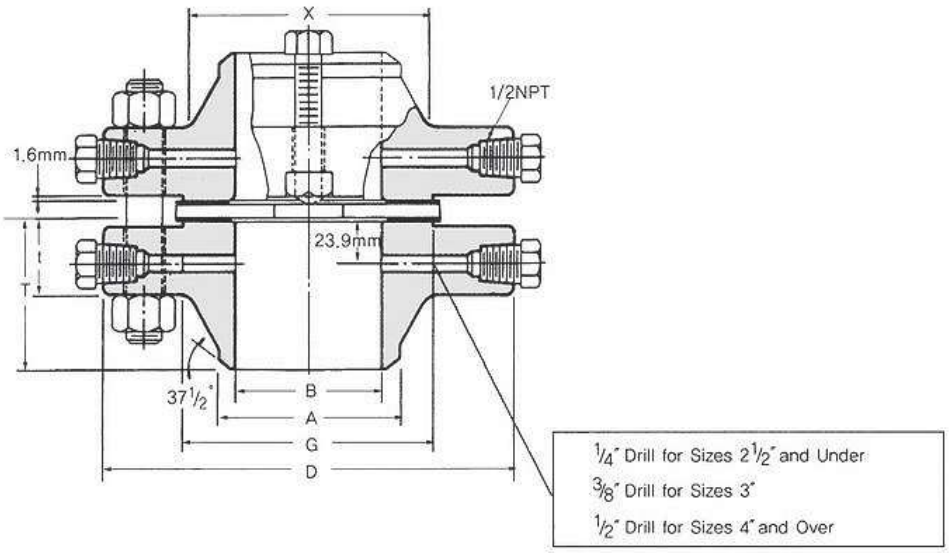
(4) Unless otherwise specified, raised face union are furnished with alloy bolt studs per ASTM A193 Grade B7 with Americn Standard heavy series hex nuts ASTM A194 Class 2H.

(5) On ring joint flanges having a groove depth 0.375"(9.5mm) and less, the distance from the center line of the tap hole to the flange face is 0.750"(19.1mm). When the depth of groove is 0.438"(11.1mm) or greater, changes in drill size or method of drilling are necessary.

(6) Bolt lengths for raised face flanges include allowance for orifice and gasket thickness of 0.25"(6.4mm) for NPS 4-12 and 0.38"(9.7mm) for NPS 14-24. Bolt lengths for ring type joint flanges include allowance of 0.62"(15.7mm) for NPS 4-10, 0.75"(19.1mm) for NPS 12-18 and 0.88"(22.4mm) for NPS 20.

CLASS 300 ORIFICE FLANGES

WELDING NECK (RAISED FACE)



ASME B16.36 FORGED FLANGES

Unit:mm

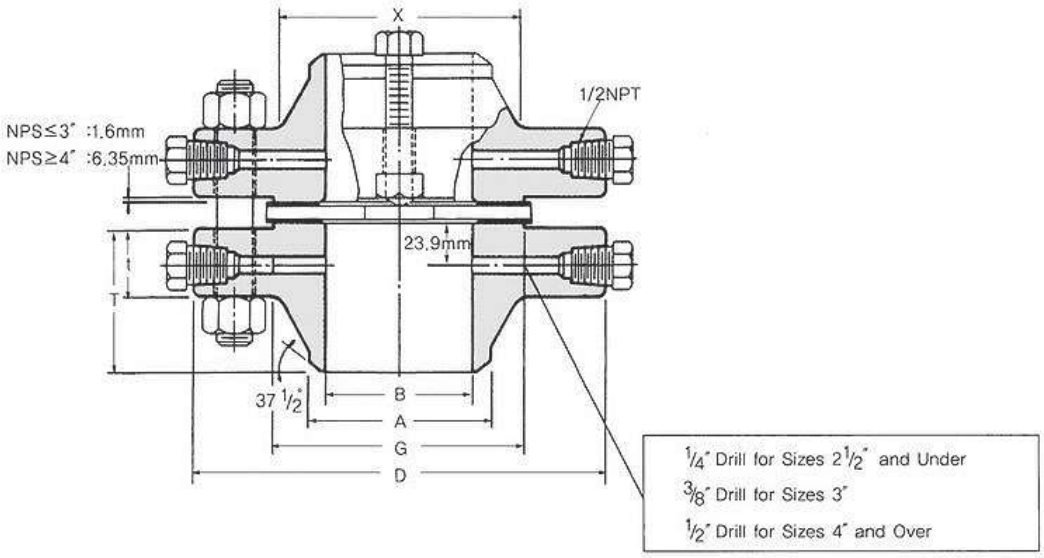
Nominal Pipe Size	Outside Diam.of Flange	THICKNESS OF FLANGE(t)	Diam. of Hub at Base	Diam. of Raised Face	Diam. of Hub at Bevel	LENGTH THRU HUB(T)		BORE(B)	
						Welding Neck	Slip-on & Threaded	Welding Neck	Slip-on
	D	Raised Face	X	G	A	Raised Face	Raised Face		
1	124	38.1	53.8	50.8	33.5	82.6	47.8	26.7	34.5
1 1/4	133	38.1	63.5	63.5	42.2	84.1	46.0	35.1	43.2
1 1/2	155	38.1	69.9	73.2	48.3	85.9	47.8	40.9	49.5
2	165	38.1	84.1	91.9	60.5	85.9	49.3	52.6	62.0
2 1/2	191	38.1	100.1	104.6	73.2	88.9	50.8	62.7	74.7
3	210	38.1	117.3	127.0	88.9	88.9	52.3	78.0	90.7
4	254	38.1	146.1	157.2	114.3	91.9	53.8	102.4	116.1
5	279	38.1	177.8	185.7	141.2	101.6	53.8	128.3	143.8
6	318	38.1	206.2	215.9	168.4	100.1	53.8	154.2	170.7
8	381	41.1	260.4	269.7	219.2	111.3	62.0	202.7	221.5
10	445	47.8	320.5	323.9	273.1	117.3	66.5	254.5	276.4
12	521	50.8	374.7	381.0	323.9	130.0	73.2	304.8	327.2
14	584	53.8	425.5	412.8	355.6	142.7	76.2	336.6	359.2
16	648	57.2	482.6	469.9	406.4	146.1	82.6	387.4	410.5
18	711	60.5	533.4	533.4	457.2	158.8	88.9	438.2	461.8
20	775	63.5	587.2	584.2	508.0	162.1	95.3	489.0	513.1
24	914	69.9	701.5	692.2	609.6	168.1	106.4	590.6	616.0

Notes

(1) For the 'Bore' (B) of Welding Neck Flanges other than Standard Wall Thickness, refer to 15.
(2) Class 300 Welding Neck Flanges of size 24 ″(609.6mm) and smaller will be bored to match Standard Wall Pipe unless otherwise specified.
(3) Class 300 Orifice flanges will be furnished with 0.06″(1.6mm) raised face, which is included in 'Thickness' (t) and 'Length through Hub'(T)
(4) Bolt lengths for raised face flanges include allowance for orifice and gasket thickness of 0.25″(6.4mm) for NPS 1-12 and 0.38″(9.7mm) for NPS 14-24.

CLASS 600 ORIFICE FLANGES

WELDING NECK (RAISED FACE)



ASME B16.36 FORGED FLANGES

Unit:mm

Nominal Pipe Size	Outside Diam.of Flange	THICKNESS OF FLANGE(f)		Diam. of Hub at Base	Diam. of Raised Face	Diam. of Hub at Bevel	LENGTH THRU HUB(T)				BORE(B)	
							Welding Neck		Slip-on & Threaded		Welding Neck	Slip-on
	D	Raised Face	Ring Joint	X	G	A	Raised Face	Ring Joint	Raised Face	Ring Joint		
1	124	38.1	31.8	53.8	50.8	33.5	82.6	76.2	47.8	41.1	See Note(1) To be specified by purchaser	34.5
1 1/4	133	38.1	31.8	63.5	63.5	42.2	84.1	77.7	46.0	39.6		43.2
1 1/2	155	38.1	31.8	69.9	73.2	48.3	85.9	79.2	47.8	41.1		49.5
2	165	38.1	31.8	84.1	91.9	60.5	85.9	79.2	49.3	42.9		62.0
2 1/2	191	38.1	31.8	100.1	104.6	73.2	88.9	82.6	50.8	44.5		74.7
3	210	38.1	31.8	117.3	127.0	88.9	88.9	82.6	52.3	46.0		90.7
4	273	38.1	38.1	152.4	157.2	114.3	101.6	101.6	53.8	53.8		116.1
5	330	44.5	44.5	189.0	185.7	141.2	114.3	114.3	60.5	60.5		143.8
6	356	47.8	47.8	222.3	215.9	168.4	117.3	117.3	66.5	66.5		170.7
8	419	55.6	55.6	273.1	269.7	219.2	133.4	133.4	76.2	76.2		221.5
10	508	63.5	63.5	342.9	323.9	273.1	152.4	152.4	85.9	85.9		276.4
12	559	66.5	66.5	400.1	381.0	323.9	155.4	155.4	91.9	91.9		327.2
14	603	69.9	69.9	431.8	412.8	355.6	165.1	165.1				
16	686	76.2	76.2	495.3	469.9	406.4	177.8	177.8				
18	743	82.6	82.6	546.1	533.4	457.2	184.2	184.2				
20	813	88.9	88.9	609.6	584.2	508.0	190.5	190.5				
24	940	101.6	101.6	717.6	692.2	609.6	203.2	203.2				

Notes

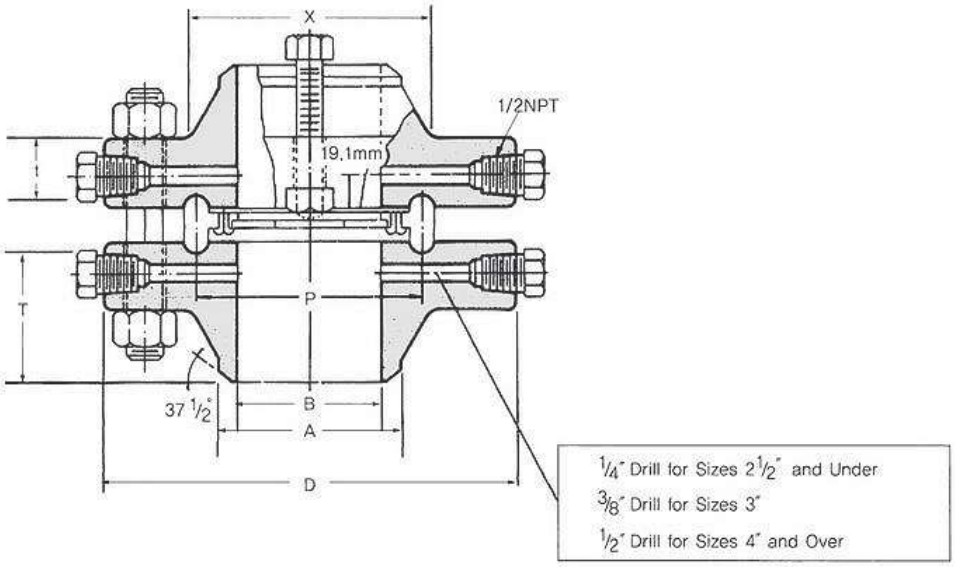
(1) For the inside diameter of pipes (corresponding to 'Bore'(B1) of Welding Neck Flanges), refer to page 15.

(2) Class 600 Flanges of sizes 3˝(76.2mm) and smaller will be furnished with 0.06˝(1.6mm) raised face, which is included in 'Thickness' (t) and 'Length through Hub'(T)
The 0.25˝(6.4mm) raised face for sizes 4˝(101.6mm) and larger is not included in (t) and (T).

(3) Each union includes two carbon steel jack screw bolts with hex nuts.

(4) Bolt lengths for raised face flanges include allowance for orifice and gasket thickness of 0.25˝(6.4mm) for NPS 1-12 and 0.38˝(9.7mm) for NPS 14-24. Bolt lengths for ring type joint flanges include allowance of 0.62˝(15.7mm) for NPS 1-10, 0.75˝(19.1mm) for NPS 12-18 and 0.88˝(22.4mm) for NPS 20.

WELDING NECK (RING–JOINT TYPE)



Unit:mm

Nominal Pipe Size	Pitch Diam.of Ring and Groove	Ring Number	DEPTH OFJACK SCREW SLOT		JACK SCREW SIZE		DRILLING TEMPLATE					
			Raised Face	Ring Joint	Raised Face (inch)	Ring Joint (inch)	Diam. of Bolt Circle	Number of Bolt	Diam. of Stud Bolts (inch)	Diam. of Bolt Holes	Length of Stud Bolts	
	P										Raised Face	Ring Joint
1	50.8	R16	9.7	6.4	5/8 X 4.00	5/8 X 4.75	88.9	4	5/8	17.5	127.0	146.1
1 1/4	60.3	R18	9.7	6.4	5/8 X 4.00	5/8 X 4.75	98.6	4	5/8	17.5	127.0	120.7
1 1/2	68.3	R20	12.7	6.4	3/4 X 4.25	3/4 X 5.00	114.3	4	3/4	21.0	133.4	152.4
2	82.6	R23	9.7	6.4	5/8 X 4.00	5/8 X 4.75	127.0	8	5/8	17.5	127.0	152.4
2 1/2	101.6	R26	12.7	6.4	3/4 X 4.25	3/4 X 5.00	149.4	8	3/4	20.6	133.4	158.8
3	123.8	R31	12.7	6.4	3/4 X 4.25	3/4 X 5.00	168.1	8	3/4	20.6	133.4	158.8
4	149.2	R37	6.4	15.7	3/4 X 3.00	3/4 X 4.00	200.2	8	7/8	25.4	139.7	152.4
5	181.0	R41	6.4	15.7	3/4 X 3.00	3/4 X 4.00	235.0	8	7/8	25.4	146.1	158.8
6	211.1	R45	12.7	22.4	1X 3.50	1X 4.00	269.7	12	7/8	25.4	158.8	165.1
8	269.9	R49	12.7	22.4	1X 3.50	1X 4.50	330.2	12	1	28.4	171.5	184.2
10	323.9	R53	12.7	22.4	1X 4.00	1X 4.50	387.4	16	1 1/8	31.8	190.5	203.2
12	381.0	R57	12.7	22.4	1X 4.00	1X 5.00	450.9	16	1 1/4	35.1	203.2	215.9
14	419.1	R61	12.7	22.4	1X 4.25	1X 5.00	514.4	20	1 1/4	35.1	209.6	228.6
16	469.9	R65	12.7	22.4	1X 4.25	1X 5.00	571.5	20	1 3/8	38.1	222.3	235.0
18	533.4	R69	12.7	22.4	1X 4.50	1X 5.00	628.7	24	1 3/8	38.1	235.0	241.3
20	584.2	R73	12.7	22.4	1X 4.75	1X 5.50	685.8	24	1 1/2	41.1	247.7	260.4
24	692.2	R77	12.7	22.4	1X 5.00	1X 6.00	812.8	24	1 3/4	47.8	279.4	285.8

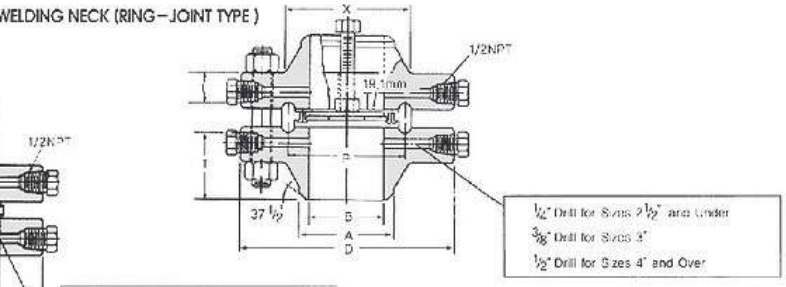
(4) Unless otherwise specified, raised face union are furnished with alloy bolt studs per ASTM A193 Grade B7 with Americn Standard heavy series hex nuts ASTM A194 Class 2H.

(5) On ring joint flanges having a groove depth 0.375"(9.5mm) and less, the distance from the center line of the tap hole to the flange face is 0.750"(19.1mm). When the depth of groove is 0.438"(11.1mm) or greater, changes in drill size or method of drilling are necessary.

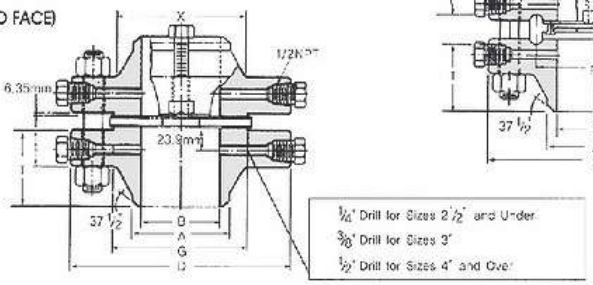
(6) Bolt lengths for raised face flanges include allowance for orifice and gasket thickness of 0.25"(6.4mm) for NPS 4-12 and 0.38"(9.7mm) for NPS 14-24. Bolt lengths for ring type joint flanges include allowance of 0.62"(15.7mm) for NPS 4-10, 0.75"(19.1mm) for NPS 12-18 and 0.88"(22.4mm) for NPS 20.

ANSI B16.5 | CLASS 1500 ORIFICE FLANGES

WELDING NECK (RING–JOINT TYPE)



WELDING NECK (RAISED FACE)



Nominal Pipe Size	Outside Diam.of Flange	THICKNESS OF FLANGE(t)		Diam. of Hub at Base	Diam. of Raised Face	Diam. of Hub at Bevel	LENGTH THRU HUB(T)				BORE(B)	
							Welding Neck		Slip-on & Threaded		Welding Neck	Slip-on
	D	Raised Face	Ring Joint	X	G	A	Raised Face	Ring Joint	Raised Face	Ring Joint		
CLASS 1500												
1	149	38.1	38.1	52.3	50.8	33.5	82.6	82.6	47.8	44.5	To be specified by purchaser	34.5
1 1/4	159	35.1	35.1	63.5	63.5	42.2	73.2	73.2	47.8	44.5		43.2
1 1/2	178	38.1	38.1	69.9	73.2	48.3	88.9	88.9	47.8	44.5		49.5
2	216	38.1	38.1	104.6	91.9	60.5	101.6	101.6	57.2	57.2		62.0
2 1/2	244	41.1	41.1	124.0	104.6	73.2	104.6	104.6	63.5	63.5		74.7
3	267	47.8	47.8	133.4	127.0	88.9	117.3	117.3	73.2	73.2		90.7
4	311	53.8	53.8	162.1	157.2	114.3	124.0	124.0	90.4	90.4		116.1
5	375	73.2	73.2	196.9	185.7	141.2	155.4	104.6	104.6	104.6		143.8
6	394	82.6	82.6	228.6	215.9	168.4	171.5	171.5	119.1	119.1		170.7
8	483	92.0	92.0	292.1	269.7	219.2	212.9	212.9	142.7	142.7		221.5
10	584	108.0	108.0	368.3	323.9	273.1	254.0	254.0	158.8	158.8		276.4
12	673	124.0	124.0	450.9	381.0	323.9	282.4	282.4	180.8	180.8		327.2
14	749	133.4		495.3	412.8	355.6	298.5					
16	826	146.1		552.5	469.9	406.4	311.2					
18	914	162.1		596.9	533.4	457.2	327.2					
20	984	177.8		641.4	584.2	508.0	355.6					
24	1168	203.2		762.0	692.2	609.6	406.4					

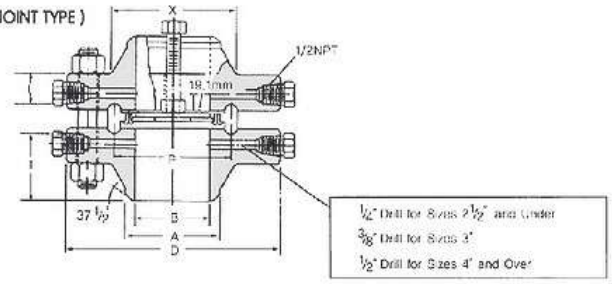
Unit:mm

Nominal Pipe Size	Pitch Diam.of Ring and Groove	Ring Number	DEPTH OFJACK SCREW SLOT		JACK SCREW SIZE		DRILLING TEMPLATE					
			Raised Face	Ring Joint	Raised Face (inch)	Ring Joint (inch)	Diam. of Bolt Circle	Number of Bolt	Diam. of Stud Bolts (inch)	Diam. of Bolt Holes	Length of Stud Bolts	
	P										Raised Face	Ring Joint
CLASS 1500												
1	50.8	R16	6.4	12.7	5/8 X 3.00	5/8 X 3.50	101.6	4	7/8	25.4	152.4	158.8
1	60.3	R18	6.4	12.7	5/8 X 3.00	5/8 X 3.50	111.3	4	7/8	25.4	139.7	146.1
1 1/4	68.3	R20	6.4	12.7	5/8 X3.00	5/8 X 3.50	124.0	4	1	28.4	158.8	165.1
2 1/2	95.3	R24	6.4	12.7	5/8 X 3.00	5/8 X 4.00	165.1	8	1 7/8	25.4	152.4	165.1
2	108.0	R27	6.4	12.7	5/8 X 3.00	5/8 X 4.00	190.5	8	1	28.4	165.1	177.8
3 1/2	136.5	R35	9.7	15.7	5/8 X 3.50	3/4 X 4.00	203.2	8	1 1/8	31.8	184.2	196.9
4	161.9	R39	9.7	15.7	3/4 X 3.50	3/4 X 4.50	241.3	8	1 1/4	35.1	203.2	215.9
5	193.7	R44	9.7	15.7	3/4 X 3.50	3/4 X 4.50	292.1	8	1 1/2	41.1	247.7	260.4
6	211.1	R46	15.8	22.4	1X6.00	1X 6.50	317.5	12	1 3/8	38.1	266.7	279.4
8	269.9	R50	15.7	22.4	1X 6.50	1X 6.50	393.7	12	1 5/8	44.5	298.5	317.5
10	323.9	R54	15.7	22.4	1X 6.50	1X 7.00	482.6	12	1 7/8	50.8	342.9	362.0
12	381.0	R58	15.7	22.4	1X 6.50	1X 8.00	571.5	16	2	53.8	381.0	406.4
14							635.0	16	2 1/4	60.5	412.8	
16							704.9	16	2 1/2	66.5	450.9	
18							774.7	16	2 3/4	73.2	501.7	
20							831.9	16	3	79.2	546.1	
24							990.6	16	3 1/2	91.9	622.3	

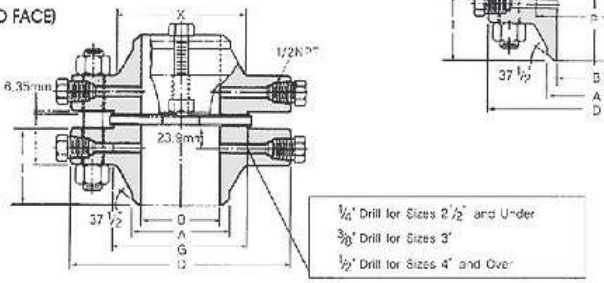
ANSI B16.5

CLASS 900 ORIFICE FLANGES

WELDING NECK (RING–JOINT TYPE)



WELDING NECK (RAISED FACE)



ASME B16.36 FORGED FLANGES

Unit:mm

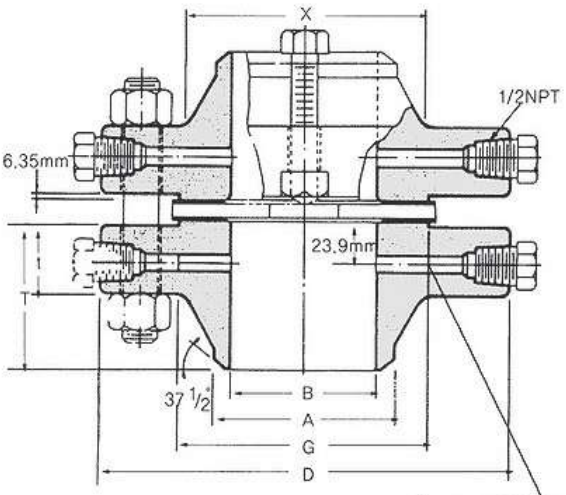
Nominal Pipe Size	Outside Diam.of Flange	THICKNESS OF FLANGE(t)		Diam. of Hub at Base	Diam. of Raised Face	Diam. of Hub at Bevel	LENGTH THRU HUB(T)				BORE(B)	
							Welding Neck		Slip-on & Threaded		Welding Neck	Slip-on
	D	Raised Face	Ring Joint	X	G	A	Raised Face	Ring Joint	Raised Face	Ring Joint		
CLASS 900												
3	241	38.1	38.1	127.0	127.0	88.9	101.6	101.6	53.8	53.8	To be specified by purchaser	90.7
4	292	44.5	44.5	158.8	157.2	114.3	114.3	114.3	69.9	69.9		116.1
5	349	50.8	50.8	190.5	185.7	141.2	127.0	127.0	79.2	79.2		143.8
6	381	55.6	55.6	235.0	215.9	168.4	139.7	139.7	85.9	85.9		170.7
8	470	63.5	63.5	298.5	269.7	219.2	162.1	162.1	101.6	101.6		221.5
10	546	69.9	69.9	368.3	323.9	273.1	184.2	184.2	108.0	108.0		276.4
12	610	79.2	79.2	419.1	381.0	323.9	200.2	200.2	117.3	117.3		327.2
14	641	85.9		450.9	412.8	355.6	212.9					
16	705	88.9		508.0	469.9	406.4	215.9					
18	787	101.6		565.2	533.4	457.2	228.6					
20	857	108.0		622.3	584.2	508.0	247.7					
24	1041	139.7		749.3	692.2	609.6	292.1					

Unit:mm

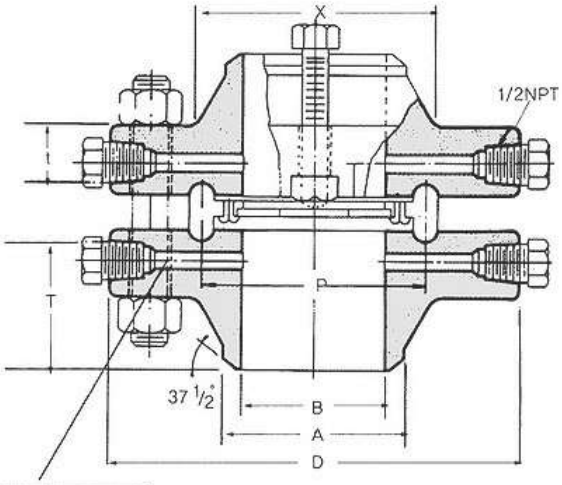
Nominal Pipe Size	Pitch Diam.of Ring and Groove	Ring Number	DEPTH OFJACK SCREW SLOT		JACK SCREW SIZE		DRILLING TEMPLATE					
			Raised Face	Ring Joint	Raised Face (inch)	Ring Joint (inch)	Diam. of Bolt Circle	Number of Bolt	Diam. of Stud Bolts (inch)	Diam. of Bolt Holes	Length of Stud Bolts	
	P										Raised Face	Ring Joint
CLASS 900												
3	123.8	R31	9.7	15.7	3/4 X 3.50	3/4 X 4.00	190.5	8	7/8	25.4	152.4	165.1
4	149.2	R37	9.7	15.7	3/4 X 3.50	3/4 X 4.50	235.0	8	1 1/8	31.8	177.8	190.5
5	181.0	R41	9.7	15.7	3/4 X3.50	3/4 X 4.50	279.4	8	1 1/4	35.1	190.5	203.2
6	211.1	R45	15.7	22.4	1X 4.50	1X 4.75	317.5	12	1 1/8	31.8	196.9	209.6
8	269.9	R49	15.7	22.4	1X 4.50	1X 5.00	393.7	12	1 3/8	38.1	228.6	241.3
10	323.9	R53	15.7	22.4	1X 4.50	1X 5.25	469.9	16	1 3/8	38.1	241.3	254.0
12	381.0	R57	15.7	22.4	1X 4.50	1X 5.50	533.4	20	1 3/8	38.1	260.4	273.1
14							558.8	20	1 1/2	41.1	279.4	
16							616.0	20	1 5/8	44.5	292.1	
18							685.8	20	1 7/8	50.8	330.2	
20							749.3	20	2	53.8	355.6	
24							901.7	20	2 1/2	66.5	444.5	

CLASS 2500 ORIFICE FLANGES

WELDING NECK (RAISED FACE)



WELDING NECK (RING–JOINT TYPE)



1/4" Drill for Sizes 2 1/2" and Under
3/8" Drill for Sizes 3"
1/2" Drill for Sizes 4" and Over

ASME B16.36 FORGED FLANGES

Unit:mm

Nominal Pipe Size	O.D. of Flange Face	O.D. of Raised Flange	Thickness of Flange	Length Thru	Diam. of Hub	Diam. of Hub at Bevel	Bore	Ring Joint Type	Ring Number	DRILLING TEMPLATE					
								Pitch Diam.		Diam. Bolt Circle	Number of Holes	Diam. of Holes	Diam. of Bolt (inch)	Length of Stud Bolts	
	D	G	t	T	X	A	B	P						Raised Face	Ring Joint
1	159	50.8	38.1	91.9	57.2	33.5	See Note(1) To be specified by purchaser	60.3	R18	108.0	4	25.4	7/8	152.4	158.8
1 1/4	203	73.2	44.5	111.3	79.2	48.3		82.6	R23	146.1	4	31.8	1 1/8	177.8	190.5
2 1/2	235	91.9	50.8	127.0	95.3	60.5		101.6	R26	171.5	8	28.4	1	184.2	196.9
2	267	104.6	57.2	142.7	114.3	73.2		111.1	R28	196.9	8	31.8	1 1/8	203.2	215.9
3 1/2	305	127.0	66.5	168.1	133.4	88.9		127.0	R32	228.6	8	35.1	1 1/4	228.6	241.3
4	356	157.2	76.2	190.5	165.1	114.3				273.1	8	41.1	1 1/2	260.4	
6	483	215.9	108.0	273.1	235.0	168.4				368.3	8	53.8	2	349.3	
8	552	269.7	127.0	317.5	304.8	219.2				438.2	12	53.8	2	387.4	
10	673	323.9	165.1	419.1	374.7	273.1				539.8	12	66.5	2 1/2	489.0	
12	762	381.0	184.2	463.6	441.5	323.9				619.3	12	73.2	2 3/4	539.8	

Notes

(1) For the inside diameter of pipes (corresponding to 'Bore' (B) of Welding Neck Flanges), refer to page 15.
(2) Class 2500 flanges will be furnished with 0.25" (6.4mm) raised face, which is not included in 'Thickness'(t) and 'Length through Hub'(T).
(3) Each union includes two carbon steel jack screw bolts with hex nuts.
(4) Unless otherwise specified, raised face union are furnished with alloy bolt studs per ASTM A193 Grade B7 with Americn Standard heavy series hex nuts ASTM A194 Class 2H.
(5) On ring joint flanges having a groove depth 0.375" (9.5mm) and less, the distance from the center line of the tap hole to the flange face is 0.750" (19.1mm). When the depth of groove is 0.438" (11.1mm) or greater, changes in drill size or method of drilling are necessary.
(6) Class 2500 Slip-on flanges are not covered by ANSI B16.5.
(7) Bolt lengths for raised face flanges include allowance for orifice and gasket thickness of 0.25" (6.4mm) for NPS 4-12 and 0.38" (9.7mm) for NPS 14-24. Bolt lengths for ring type joint flanges include allowance of 0.62" (15.7mm) for NPS 4-10, 0.75" (19.1mm) for NPS 12-18 and 0.88" (22.4mm) for NPS 20.